

Course Details & Expectations

AGST 50104 Experimental Design

Dr. Samuel B Fernandes

2026-01-01

Learning Objectives

By the end of this course, you will:

1. **Design** rigorous experiments (CRD, RCBD, factorial, split-plot, etc.)
2. **Choose** appropriate designs based on constraints and research questions
3. **Analyze** data using ANOVA, linear mixed models, and GLMs
4. **Interpret** statistical results in practical, domain-relevant language
5. **Communicate** results clearly (reports, presentations, visualizations)
6. **Recognize** and avoid common pitfalls (confounding, pseudoreplication, incorrect error terms)
7. **Use R & Quarto** for reproducible, professional analyses

Timing: ~3 minutes

Motivation: “These are the tools and concepts that professional researchers use every day.”

Course Structure & Grading

Grade Components (1000 points):

Component	Points	%
Homework	300	30%
Exam I	150	15%
Exam II	150	15%
Final Exam	200	20%
Project	200	20%

Letter Grade Scale:

Grade	Range	%
A	900–1000	90%+
B	800–899	80%+
C	700–799	70%+
D	600–699	60%+
F	0–599	<60%

Timing: ~2 minutes

Emphasis: Exams are doable; projects are meaningful.

Homework & Teamwork

How Homework Works:

- Groups of ~**5 students** (formed Week 1)
- Assigned almost every **Friday**
- **Due Thursday 11:59 PM** (written submission)
- **Friday Lab:** One member randomly selected to present
- Lowest homework score is **dropped** at end of semester

Grading is 60% group + 20% your presentation + 20% peer evaluation

Your individual grade depends on effort AND how you're perceived by teammates.

Team Success Tips:

- Communicate early and often
- Divide work fairly
- Submit **Thursday, not Friday**
- Practice explaining your work
- Use AI (GitHub Copilot) to learn faster
- Office hours are for team troubleshooting

Timing: ~4 minutes

Story: “Group work teaches you real research—you’ll do this in industry or grad school. Start strong!”

Exams: What to Expect

Caution

AI Policy on Exams:

- **NO AI during exams** (GitHub Copilot, ChatGPT, etc.)
- Unauthorized use = Academic integrity violation
- Exams test **YOUR** understanding, not AI's

Exams are a fair assessment of your knowledge.

- **Exam I** (Feb 20): CRD through early RCBD concepts
- **Exam II** (April 3): Factorial to Incomplete Block designs
- **Final Exam** (May 6): Comprehensive; can include anything from the course
- **Format:** Closed-book; bring **one handwritten formula sheet** (single side)
- **Exam can be handwritten OR coded** (we'll practice both)

Timing: ~4 minutes

Emphasize: Exams are designed to be fair. If you attend labs and complete homework, you'll be prepared.

The Final Project

Project Requirements:

- **Topic:** Randomly assigned in Week 1 (one of 9–11 experimental designs)
- **Format:** 9–11 minute video presentation (unlisted YouTube)
- **Deadline:** April 24, 2026, 11:59 PM
- **Grading:** 200 points = 50% concept explanation + 50% applied example
- **All group members must appear** in the video to receive a grade

Projects showcase REAL experimental design thinking—not just formulas!

What to Include:

- Clear explanation of your assigned design
- Why it's appropriate for the scenario
- Complete analysis workflow:
 - Data exploration
 - Statistical analysis
 - Diagnostics

– Conclusions

- Visualizations that tell the story

Timing: ~4 minutes

Guidance: “Your project is a portfolio piece. Make it polished!”

Computing & Resources

Required Software (Free):

1. **R** (statistical computing)
2. **VS Code** (code editor + Quarto)
3. **Quarto** (reproducible documents)
4. **GitHub Account** (version control + Copilot)
5. **GitHub Copilot** (AI programming assistance)

Alternative: Posit Cloud (free online R/Quarto)

Key Packages You’ll Learn:

- **dplyr**: Data manipulation
- **ggplot2**: Visualization
- **tidyr**: Data reshaping
- **car**: ANOVA & diagnostics
- **emmeans**: Post-hoc comparisons
- **lme4/nlme**: Mixed models

Install these as we go—no upfront burden!

GitHub Copilot helps you write R code faster. It’s built into VS Code and free with your .edu email.

Timing: ~3 minutes

Encouragement: “Don’t worry about installing everything now. We’ll set it up together in Week 1 Lab.”

Course Expectations & Policies

- **Attendance:** Expected at all lectures and labs (excused absences: illness, family, religious observance, etc.)
- **Office Hours:** Monday & Wednesday, 1–2 PM (PTSC 107), or by appointment
- **Communication:** Please allow 24 hrs (weekdays) or 48–72 hrs (weekends) for email responses
- **Procrastination:** Homework is due Thursday 11:59 PM—don’t start Friday morning!
- **Respect:** This is a collaborative environment. Support your teammates and classmates

Timing: ~2 minutes

Tone: Professional but supportive.

Learning Objectives

Reminder: We reviewed these outcomes at the start of class.

Questions?

Contact Information:

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Office Hours:

Mon & Wed: 1:00–2:00 PM
Or by appointment

Course Materials:

- Lectures: GitHub Pages
- Assignments: Blackboard Ultra
- Communication: Microsoft Teams (per group)
- Announcements: Email + Blackboard

Everything is linked in Blackboard!

Welcome to the team! Let’s design some amazing experiments together.

Timing: ~2 minutes

Closing: “Reach out early if you have questions or concerns. I’m here to help you succeed.”